Real-time keyword spotting on ARM Cortex-M4F

Build and test

Continuous voice control on an **Arduino Nano 33 BLE Sense / nRF52840**, running **Arduino Zephyr Boards 1.0.0**. The microphone captures mono audio at 16 kHz, CMSIS-DSP extracts log-mel features, and an int8 TensorFlow Lite Micro model scores the keyword **"yes."** Confirmed detections toggle the built-in LED. The application runs continuously while powered, independently of a serial monitor.

Zephyr is the sole build target. Two C++ application threads separate audio processing from device control. The core's PDM thread acquires samples; it can preempt inference, preserving acquisition while the audio thread spends roughly 55 ms on a model invocation. All threads share one 64 MHz Cortex-M4F CPU.

Build, test, and upload

Requirements: Git, Make, Python 3.9+, HTTPS access for setup, and a host c++ compiler with C++20 and AddressSanitizer/UndefinedBehaviorSanitizer support. macOS Command Line Tools and Ubuntu 24.04 are suitable host environments. The cross-compiler is installed by Arduino CLI, not supplied by the host compiler.

```
make setup                          # Install locked tools and libraries locally
make test-unit                      # Isolated component tests
make test-regression                # Units plus integration and build-tool checks
make test-pdm-driver                # Actual pinned driver fault test; needs setup
make build JOBS=4                   # Compile for Zephyr
make boards                         # Find the connected Nano
make flash PORT=/dev/cu.usbmodem11301 JOBS=4
```

Substitute your actual USB port; Linux commonly uses `/dev/ttyACM0`. Close any other serial monitor before uploading. Arduino CLI installs/updates the official Zephyr loader automatically. Double-press RESET if bootloader recovery is needed. On Apple Silicon, the packaged Intel uploader requires Rosetta 2.

All tools, downloads, libraries, staged sources, objects, and binaries stay under ignored `.build/`. Existing global Arduino installations are not used. Edit `firmware/`, never the generated sketch. Outputs and their exact input manifest are in `.build/zephyr-artifacts/`. The selected FQBN is `arduino:zephyr_main:nano33ble:link_mode=static`.

The locked inputs are Arduino CLI 1.5.1, Arduino Zephyr Boards 1.0.0, its official `arm-zephyr-eabi@1.0.1` compiler (GCC 14.3.0), Arduino_CMSIS-DSP 5.7.0, and the exact TensorFlow Lite source revision recorded in dependencies.lock.json. Firmware and host tests use C++20. See build and CI for provenance, prerequisites, optional IDE settings, troubleshooting, and PDF generation.

Runtime architecture and ownership

From	Connection	To
16 kHz PDM microphone	Flow	Core PDM thread: priority 3
Core PDM thread: priority 3	Flow	8192-sample SPSC ring
Core PDM thread: priority 3	binary semaphore	Audio thread: priority 4
8192-sample SPSC ring	Flow	Audio thread: priority 4
Audio thread: priority 4	Flow	512-point FFT / 20 log-mel bands
512-point FFT / 20 log-mel bands	Flow	61 x 20 int8 context / inference
61 x 20 int8 context / inference	Flow	Zephyr queue: 8 score values
Zephyr queue: 8 score values	Flow	Control thread: priority 5
Control thread: priority 5	Flow	Freshness / confidence filter / LED
Control thread: priority 5	metrics copy	Main thread: priority 6
Audio thread: priority 4	metrics copy	Main thread: priority 6
Main thread: priority 6	Flow	Best-effort USB telemetry

Lower Zephyr priority numbers run first. The audio thread owns feature extraction, the model, and stream recovery. Control owns temporal detection and the LED. The lower-priority main thread owns telemetry, isolating internal Serial waits from keyword control. A queue copies score/timestamp/sequence/generation values, never pointers into shared audio or tensor buffers. A delayed control thread can lose scores without blocking the audio consumer. An overflowing audio ring rejects new samples and explicitly resets detection evidence.

Module	Responsibility
`kws_pipeline`	Zephyr threads, fixed stacks, semaphore, score queue, metrics mutex, initialization and cleanup
`audio_capture` / `audio_queue`	PDM callback routing, bounded reads, lock-free sample publication, complete-hop reads and loss counters
`audio_pipeline`	DSP readiness, backlog-aware inference, generation reset and timing
`feature_extractor` / `fft_power`	PCM scaling, frame overlap, periodic Hann, packed real FFT, power, mel/log and int8 features
`keyword_model`	Single interpreter, four operator types, fixed 64 KiB tensor arena and score decoding
`keyword_event` / `keyword_filter`	Freshness, sequence continuity, confidence smoothing, hysteresis and cooldown
`device_control` / `telemetry`	LED action and separate best-effort diagnostic output

The application and model runtime have static lifetime. Fixed application stacks reserve 8192 bytes for audio and 6144 for control. A nonblocking mutex protects only short metrics copies. Zephyr IRQ keys preserve nested capture snapshots; scheduler locking covers the final freshness check and initialized GPIO write while leaving interrupts enabled. See RTOS design and C++ ownership. Application buffers and kernel resources use fixed storage, including the inference arena. The build rejects direct heap allocation/deallocation imports in sketch objects, disables exceptions/RTTI and rejects variable-length arrays. Each audio activation handles at most eight hops; a full batch yields

for 1 ms before resuming. These controls do not establish a whole-system WCET or prove that loader/driver code never allocates.

The build also stages a PDM driver patch that sleeps 1 ms before retrying a failed device read. Run `make test-pdm-driver` after setup to fault-test the actual worker; CI runs this additional gate. The installed core and official compiler/loader remain intact.

Features, model, and detection

Each 256-sample hop advances a 512-sample frame with 50% overlap. The extractor uses a periodic Hann window, 257 correctly unpacked FFT power bins, a 20-band mel matrix spanning 20-4000 Hz, natural log with epsilon 1e-6, and model-supplied int8 quantization. The 61-row context covers 992 ms of audio. First inference requires a full context; subsequent attempts require eight new rows and a drained hop backlog, giving a nominal 128 ms decision interval.

The deployed 8496-byte model classifies `yes`, unknown speech, and silence. Its six convolution operators contain 992 int8 weights and 83 int32 biases: **1075 deployed parameter values**, or 1324 parameter payload bytes. Its eight operators use Conv2D, DepthwiseConv2D, MEAN and SOFTMAX registrations. The model and mel coefficients are compiled into flash; the 64 KiB arena holds mutable inference data. Startup reports actual tensor allocation.

Control uses class-0 confidence, EMA weight 0.5, raw/smoothed thresholds 0.8/0.75, two qualifying decisions, release below 0.4, and a 1500 ms cooldown. Events expire after 250 ms; sequence/generation gaps discard partial evidence while retaining latch and cooldown state. A final scheduler-locked check prevents an obsolete score from triggering an action after audio invalidates its generation.

See DSP details for formulas, packed FFT layout and training alignment, and TinyML inference for the exact model graph, quantization, training results, inspection commands and accuracy limits.

Measured Zephyr results

The following observations come from the October 22, 2026 Zephyr hardware session. They describe the recorded image and workload, not worst-case guarantees. The cleanup build and any new measurements are identified separately in validation.

Measurement	Recorded result
Sampling rate over normal capture	Approximately 15,999.7 samples/second
Normal DSP average / observed maximum	2.217 / 5.249 ms per consumed hop
Normal inference average / observed maximum	54.994 / 55.054 ms
Normal processing-step maximum	57.342 ms
Audio-ring high-water mark	768 of 8192 samples
Normal losses and errors	Zero reported sample losses, missed buffers, inference errors, notification errors, event drops or stale events
Spoken keyword check	16 cumulative confirmed actions; user confirmed physical LED toggling on "yes"
Deliberate 1500 ms control stall	Four expected event drops, nine stale scores rejected, no reported audio loss or inference errors
Recorded sketch flash / RAM	206,032 / 118,096 bytes
Stock loader sketch RAM region	131,072 bytes; recorded remainder 12,976 bytes

Measurement	Recorded result
Tensor arena reserved / measured used	65,536 / 12,500 bytes

The RAM figure already includes both application stacks, ring, DSP workspace, and reserved arena. The remainder is not measured free heap. Stock loader stack watermarks are unavailable: `stack_free=0/0` means unmeasured. General system heap, framework/interrupt stacks, extended endurance, cold power-cycle, and calibrated acoustic accuracy remain separate validation work. See real-time limits.

The table preserves the original image's evidence. The later C++20 hardening build uses 206,824 bytes flash. Current validation records its source hashes, captures and limits; speech/LED was not repeated.

The saved training notebook reports 93.48% floating-point held-out test accuracy, 97.90% validation accuracy, and 98.02% final training accuracy. These are training artifacts, not measured recognition rates for the quantized model on this board. Speech/LED checks establish functionality without an accuracy denominator.

Hardware checks and telemetry

Install PySerial in a local environment, then capture normal operation:

```
python3 -m venv .build/hardware-venv
.build/hardware-venv/bin/python -m pip install pyserial==3.5
.build/hardware-venv/bin/python tests/hardware_capture.py --port /dev/cu.usbmodem11301 \
    --seconds 60 --allow-unavailable-stack-metrics --output .build/normal.json
.build/hardware-venv/bin/python scripts/rtos_stress.py --port /dev/cu.usbmodem11301 \
    --seconds 25 --output .build/stress.json
```

The stress script uploads a test-only control-stall image and restores the normal image in `finally`; confirm restoration succeeds. Speak "yes" with pauses and observe the LED separately. Ending a capture leaves the firmware running.

Telemetry field	Meaning
`ms`, `samples`	Uptime and cumulative samples received by the callback
`dropped`, `missed_buffers`	Rejected ring samples and callbacks with lost/invalid reads
`queue_max`, `gap_max_us`, `audio_age_us`	Ring high-water mark, maximum callback gap, age of last received audio
`dsp_us`, `infer_us`, `process_us`	Average / observed maximum DSP, inference and one pipeline step
`response_us`	Last / maximum response from estimated first high-confidence window end to action; not acoustic-onset latency
`inferences`, `errors`, `dsp_errors`, `p_milli`	Invocation count, inference/DSP failures and latest class-0 score scaled by 1000
`triggers`, `led`	Confirmed action count and logical LED state
`event_drops`, `stale_events`	Dropped classification results and rejected expired/old-generation results
`notify_errors`	Reserved diagnostic counter; Zephyr binary-semaphore give has no failure return
`event_us`	Average / maximum inference-completion to control-validation delay
`stack_free`	Audio/control unused stack bytes if loader support exists; stock loader reports zero/zero

Counters are cumulative and may precede a host recording. Compare successive records to infer activity within an interval. A buffered telemetry line may precede the fresh lines received when reopening a monitor.

Tests and documentation

`make test` runs the same complete host gate as GitHub Actions: twelve unit entries and four additional regression entries. `make test-pdm-driver` is a separate gate after setup and also runs in firmware CI. These suites exercise actual Zephyr application modules with deterministic PDM, IRQ, serial, DSP, model and kernel fixtures. Hardware checks cover the real microphone/DSP/model path. Testing describes coverage and limits.

Line/branch coverage percentages are not measured. Fixtures do not exhaustively reproduce thread interleavings or execute the real CMSIS-DSP/TFLite numerical path; feature parity, recognition accuracy and long-duration timing need separate evidence.

Documentation index links setup, architecture, DSP, inference, ownership, real-time behavior and validation evidence. README.pdf is generated from this root README; CI checks that the export is current. Its architecture table retains every node and connection in the Markdown diagram. The other topic guides are maintained as Markdown without PDF counterparts. Model assets and the training notebook remain available under `models/` and `training/`.